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# Sustainable cyberbullying detection with category-maximized relevance of harmful phrases and double-filtered automatic optimization

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ABSTRACT

We developed a supporting solution for "cyberbullying" prevention based on recent discoveries in Artificial Intelligence and Natural Language Processing. Cyberbullying, defined as using the Internet to humiliate and slander other people has become a serious problem. In Japan members of the Parent–Teacher Association manually perform Web monitoring to stop cyberbullying activities. Unfortunately, reading through the whole Web manually is an impossible task. Although the complexity of cyberbullying makes it a problem unsolvable solely with the help of technology, we found that technology could make cyberbullying prevention more efficient. We developed a novel method of automatic detection of cyberbullying entries on the Internet. In the method we use seed words from three categories to calculate a semantic orientation score and then maximize the relevance of categories. The proposed method outperformed baseline settings in both laboratory and real world conditions. The developed system was deployed and tested in practice. After a year of testing we noticed a greater than 30 percent-point-drop in its performance. We hypothesize on the reasons for the drop. To regain the lost performance and retain it in the future we propose additional improvements including automatic acquisition and filtering of seed words. Experimentally selected optimal improvements regained much of the lost performance.

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## 1. Introduction

"Cyberbullying" (later shortened to **CB**) is a new form of bullying. It is carried out on the Internet instead of classrooms. It takes the form of hate messages sent through e-mails, Social Networking Services (SNS, such as Twitter or Facebook), or electronic Bulletin Board Systems (BBS), with the use of personal computers or mobile phones. Recently it has become a serious social problem in many countries.

In the most recent cross-cultural study summarizing the problem strongly highlighted that although CB has been appearing in various age and social groups, is the most prevalent among mostly young school pupils and students [1]. In particular, it is estimated that at least about 24% of pupils are bullied with the use of modern technology (computer, smart-phones, etc.) and Internet services such as Facebook and Twitter, and about 17% participated actively in such online bullying activities.

As more children have easier access to modern technologies and Internet nearly every day [2], this makes cyberbullying one of the most, if not the most crucial problem when it comes to children interacting with computers. Thus the problem is very important and relevant to the field of Child Computer Interaction [3]. Especially in Japan, which is the focus of this paper, the ratio of everyday use of mobile phones among schoolchildren is an astonishing 75.2% for primary school, 95.7% for junior high school, and 99.4% for senior high school [4].

Examples of CB that have actually happened, include ridiculing a student's personality, body type or appearance, on informal school BBS, slandering students or insinuating they had performed deviant sexual practices. Some cases of cyberbullying lead the bullied students to self-mutilation, even suicide, or on the other

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**Table 1**
Examples of harmful entries which were requested for deletion. Japanese (above), transliteration (middle), English translation (below). *Mobage* and *Gree* are online game service Web sites.

| |
|---|
| _ 調子乗りすぎいっぺん殺らなあかんで<br>(*Chōshi nori sugi ippen yara na akan de*)<br>"Don't get excited that much or I'll kill ya!" |
| _ 新田キモイつかキショイほんま死んで<br>(*Nitta kimoi tsuka kishoi honma shinde*)<br>"Nitta [proper noun], you're ugly, or rather fugly, just die, man" |
| _ ンな奴どつき回したれ<br>(*N'na yatsu dotsuki mawashi tare*)<br>"What an ass, slap him" |
| _ 性格わるーい　ぶちゃいく一笑<br>(*Seikaku waru–i buchaiku– warai*)<br>"Baaad personality, and an ugly hag, lol" |
| _ あの女、昔、モバだったかグリに登録してたヤリマンじゃん。<br>(*Ano onna, mukashi, Moba dattaka Guri ni tōroku shiteta yariman jan*)<br>">>?17 that woman is the same one who was bitching around before on *Mobage* or *Gree*". |
| _ すぐにヤレる。01234567890。めっちゃカワイイで<br>(*Sugu ni yareru. 01234567890. Meccha kawaii de.*)<br>"You can take her out even now. 01234567890. She's a great lay". |

hand, cause them to attack the students who put the harmful entries on the Internet.

In Japan [5], bullying in general (including cyberbullying) is considered one of the most common causes of suicide among school pupils [6,7].

Due to the severity of the problem, Japan has in recent years started a number of projects to increase the recognition of bullying in schools among both students and teachers. This caused a slow decrease in the number of reported bullying cases.[1] Unfortunately, the most disturbing is the fact that, although the number of cases reported in schools is dropping, the number of children claiming they have been bullied in fact remains the same [6,7,2]. This suggests that most of the bullying has moved online and gradually takes the form of cyberbullying.

To deal with the problem in Japan, members of Parent–Teacher Association (PTA)[2] perform Web site monitoring activities called "Internet-patrol" (or shortly "net-patrol"). When a harmful entry is detected a net-patrol member who found it sends a request to the Internet provider or Web site administrator to remove the entry. Some of the actual examples of harmful entries which were requested for deletion are represented in Table 1 (names, phone numbers and other personal information were changed).

Unfortunately, net-patrol has been mostly carried out manually. It takes much time and effort to find harmful entries (entries that contain information and expressions aimed to harm other users) in a large amount of contents appearing on a countless number of electronic Bulletin Board (BBS) pages. Moreover, the task comes with a great psychological burden to the net-patrol members. To solve this problem and decrease the burden of net-patrol members, Matsuba et al. (2011) [9] proposed a preliminary method to detect harmful entries automatically. In their method they extended the method of relevance calculation PMI-IR, developed by Turney (2002) [10] to calculate relevance of a document with harmful contents. With the use of a small number of seed words they were able to detect large numbers of document candidates containing harmful entries with satisfying effectiveness.

In particular, their method found harmful entries with an accuracy of 83% on test data for which about half contained harmful entries. However, it was not verified how well the method would perform in real life conditions, where cyberbullying entries and normal contents do not appear in equal numbers.

In this research, initially introduced by Nitta et al. in 2013 [11], we firstly propose an optimization of Matsuba et al.'s method by maximizing the relevance scores. In our method we divide the seed words into multiple categories and calculate a maximal relevance value for each seed word from each category. By calculating the maximized score, representing semantic orientation of *harmfulness* of a sentence or an entry, the method detects harmful entries more effectively than in the previous research. Moreover, we evaluated our method on data sets with real-life ratios of harmful contents to verify the usability of the method in the most realistic way.

After confirming the performance, the method was deployed and tested "in the field" for over one year. Unfortunately, after re-evaluating the method after that time we found out that the performance significantly dropped by over 30-percentage-points when compared to its original performance measured in 2013. In the original evaluation process, applied also in this research and previously in Matsuba et al. [9], Nitta et al. [11] we assigned the harmfulness score to all sentences from their dataset containing harmful and non-harmful entries, sorted all entries decreasingly and checked the performance looking at the top $n$ entries, while increasing the number of $n$ (further referred to as "threshold window") by 50 entries at a time. This way we were able to find the fluctuation in performance when more data is applied in the evaluation. For the evaluation measures we used standard Precision and Recall (Precision and Recall is explained in detail in Section 4.3). In particular, the originally confirmed high performance was on a level of 91% when it comes to Precision for low Recall values. Moreover, original performance rates from late 2013 were retained high (around 70%) even close to 50% of Recall, after which they decreased due to a drop in Precision for higher thresholds. However, when the evaluation experiment was repeated in early 2015, the overall results dropped greatly. Precision did not exceed 60% for the whole threshold span. We introduced a number of improvements to regain the lost performance and assure retaining it at a sufficient level.

We describe all the research according to the following outline. Firstly, in Section 2 we introduce the problem of cyberbullying and describe research related to the detection of cyberbullying and extraction of harmful entries form the Internet.

[1] However, it must be noticed that the actual number of cases is estimated as over three times larger [8].

[2] An organization composed of parents, teachers and school personnel aiming to improve the school experience for children.

Next, we describe the proposed method in Section 3. Furthermore, in Section 4 we construct evaluation data sets with different ratios of cyberbullying entries, perform evaluation experiments based on these data sets, and describe the results of those experiments. We also present a more detailed discussion of the results in Section 4.5. Next, we present the re-evaluation of the method after one year. We discuss the potential reasons for the drop in performance (Section 5.1), and then propose an optimization method to regain some of the performance. In particular we propose a method for automatic acquisition of seed words (Section 6) not only to improve the method performance, but also to allow sustaining the performance for future use of the method. We conducted an additional series of experiments with the automatically obtained different seed word sets and experimentally choose the one with the best performance (Section 7). Based on the analysis of functions and results of the proposed method we discuss the potential as well as limitations of the proposed method in Section 8. Finally, in Section 9 we conclude the paper and propose ideas for further improvement and future applications of the method.

## 2. Background

### 2.1. Cyberbullying—A social problem

The problem of harmful and offending messages on the Internet has existed for many years. One of the reasons such activities evolved was the anonymity of communication on the Internet, giving users the feeling that anything can go unpunished. Recently the problem has been officially defined and labeled as **cyberbullying** (CB).

The National Crime Prevention Council in the USA, defines CB as "when the Internet, cell phones or other devices are used to send or post text or images intended to hurt or embarrass another person".[3]

Although the presence of online aggression has existed for many years, most probably from the very beginning of using the Internet as a communication tool, only since about ten to fifteen years ago has it been noted as a serious social problem. During this time cyberbullying as a social problem has been studied in social sciences by a number of researchers.

Some of the first robust research on CB was done by Hinduja and Patchin, who performed numerous surveys about the subject in the USA [12,8]. They found out that the harmful information may include threats, sexual remarks, pejorative labels, or false statements aimed to humiliate others. When posted on a social network, such as Facebook or Twitter, it may disclose humiliating personal data of the victim defaming and ridiculing them personally.

Around the years 2009–2011 a number of large scale questionnaire studies and social campaigns have been conducted to measure the degree of cyberbullying and work out methods to mitigate the problem.

For example, Cross et al. (2009) in Australia [13] found out that cyberbullying in Australian schools happens for about five to eight percent of children depending on their school year. Comparable results have been noticed in the United States [14], Finland [15], and across Europe [16]. Hasenbrink et al. (2009) [17] present even larger estimates, that even one in five young people (not limited to school environment) could had experienced bullying or harassment through the Internet or mobile devices. These estimates are also confirmed in Qing (2007) [18], Pyżalski (2012) [19], and more recently by Kann et al. (2014) [20].

Cyberbullying has also been thoroughly studied and analyzed by Dooley, Pyżalski, and Cross (2009) [21], who performed an in-depth comparative analysis of traditional face-to-face bullying and cyberbullying, while Lazuras et al. (2012) [22] discussed implications of cyberbullying for teachers in school environments. Dooley, Pyżalski, and Cross (2009) point out some of the similarities, between traditional face-to-face bullying, but also mention some of the differences making cyberbullying in some aspects a problem more difficult to contain. As for the similarities, which contribute to classifying the problem as a kind of bullying, are such features as: *peer group*, such as classmates in face-to-face bullying and friends from groups on Social Networking portals (Facebook, Twitter, etc.), which in reality often overlap, and play a crucial role in victimization of bullying targets; *repetitiveness* of bullying acts, which especially with the use of Internet can occur more often than face-to-face bullying; or *imbalance of power*, meaning that usually one person, or a small group of people become bullied by an overwhelming number of bullies. This is also a feature distinguishing bullying from other types of cyber-aggression.

However, the environment and the tools used also make cyberbullying even more of a humiliating experience than its face-to-face counterpart. For example, with the use of the Internet, cyberbullying can occur on a much larger scale making it an overwhelming experience. Also with the popularity of Social Network Services, humiliating a person publicly, such as on their official Facebook homepage, makes the attack public not only for a limited number of viewers, but open to the general public, which in worst cases could even appear in search engine results, thus making a person exposed theoretically to the whole world. Additionally, the indirectness of the relationship caused by the limitations of the communication channel in the online environment, which causes emotional disconnection in some people, add to the higher possibility of becoming a bystander, or a person, who is a witness of a bullying act, but does not react. This result increases the sense of impunity in potential bullies.

In Japan, where a person's belonging and the role played in the society is considered a major defining feature of the person in general, the problem of cyberbullying and traditional bullying, often realized in the form of gossip or ostracism [23], has profound implications on one's personality and mental health. Despite this fact, research on cyberbullying in Japan has gained speed only recently [24], after several cases of suicides of CB victims not being able to bare the humiliation. The problem itself however, has been noticed earlier. In 2008, MEXT considered the problem of cyberbullying serious enough to begin action to prevent it. In a manual for handling CB cases [5], the Ministry puts a great importance on early spotting of suspicious entries, especially on Social Networking Services (SNS) and informal school Websites. The latter are Web sites where students and pupils exchange information about subjects or contents of tests, etc. However, it was noticed that such pages witnessed a rapid increase of CB toward pupils and even teachers [25].

The Internet Patrol (net-patrol) was founded to deal with the problem. Its participants are usually teachers and PTA members. Based on the MEXT definition of CB, they read through all Internet contents, and when they find a harmful entry they send a deletion request to the Web page administrator and report about the event to the Police.

Unfortunately, at present net-patrol is performed manually mostly as a voluntary activity. This includes reading the countless entries, deciding about their harmfulness, printing out or taking photos of the pages, sending deletion requests and reports to appropriate organs. The surveillance of the whole Web is an impossible task for the small number of net-patrol members. Moreover, the task comes with great psychological burden on mental health to the net-patrol members. With this research we developed and tested a tool allowing automatic detection of CB on the Internet to ease the burden of net-patrol volunteers.

[3] http://www.ncpc.org/cyberbullying.

### 2.2. Related research

Although the problem of cyberbullying has been studied in social sciences and child psychology for over ten years, there has been only few attempts to study the problem using methods from the field of Artificial Intelligence or Natural Language Processing.

Some of such research focused on extracting harmful information from the Internet. For example, Ishisaka and Yamamoto (2010) [26] have focused on developing an abusive expression dictionary based on a large Japanese electronic bulletin board "2 channel". In their research they firstly defined words and paragraphs in which the speaker directly insults or slanders other people with the use of explicit words and phrases such as バカ (*baka*) "stupid", or マスゴミノクズ (*masugomi no kuzu*) "trash of mass-mudia". Next, they studied the use of abusive language, in particular, which words appear more often with abusive expressions, and based on this study they extracted abusive expressions from the surrounding context.

Ptaszynski et al. (2010) [27] performed affect analysis of small dataset of cyberbullying entries to find out that distinctive features for cyberbullying were vulgar words. They applied a lexicon of such words to train a classifier based on Support Vector Machines (SVM). With a number of optimizations (word normalization, etc.) the system was able to detect cyberbullying with 88.2% of F-score. However, increasing the data caused a decrease in results, which made them conclude that SVMs are not ideal in dealing with frequent language ambiguities typical for cyberbullying.

In other research, Ikeda and Yanagihara (2010) [28] have manually collected and divided separate sentences into harmful and non-harmful, and based on word occurrence within the corpus they created a list of keywords for classification of harmful contents. Next, they utilized context of dependency structures of sentences containing harmful and non-harmful contents to improve the system performance. However, they could not overcome the problem of numerous variations of the same expressions on the Web, differing with only one or two characters, such as 爆破 (bakuha) "blow up" and 爆一破 (*baku–ha*) "blo–ow up". The weakness of this method is that all of the variations of the same expression need to be collected manually, which is very time-consuming.

Fujii et al. (2010) [29] proposed a system for detecting documents containing excessive sexual descriptions using a distance between two words in a sentence. In their method they determined as harmful those words which are in closer distance to the words appearing only in sexual context ("black words") rather than those in closer distance to words which appear in both sexual and non-sexual context ("grey words").

Hashimoto et al. (2010) [30] proposed a method for detecting harmful meaning in jargon. In their method they assumed that the non-standard meaning is determined by the words surrounding the word in question. They detected the harmful meaning based on calculating co-occurrence of a word with its surrounding words.

Matsuba et al. (2011) [9] proposed a method to automatically detect harmful entries, in which they extended the SO-PMI-IR score [10] to calculate relevance of a document with harmful contents. With the use of a small number of seed words they were able to detect large numbers of candidates for harmful documents with an accuracy of 83% on test data.

Nitta et al. (2013) [11] proposed an improvement to Matsuba et al.'s method. They used seed words from three categories (abusive, violent, obscene) to calculate SO-PMI-IR score and maximized the relevance of categories. Their method achieved over 90% of Precision for 10% Recall. Finally, Cano et al. (2013) [31] used weak supervision to detect generally perceived violence in social media.

Most of the previous research assumed that using vulgar words as seeds will help detecting cyberbullying. However, all of them notice that vulgar words are only one kind of distinctive vocabulary and do not cover all cases. In our research we did not focus on the specific words selected manually by expert annotators, as in previous research. Instead, we determined the harmfulness of input by calculating the harmfulness score for all applicable word sequences in input. Moreover, since we checked the co-occurrence of word sequences on the Web, our method greatly reduces the cost of manual construction of training data. Furthermore, in calculating the harmfulness score we apply dependency relations between phrases. Therefore there is no need to check all words proceeding and succeeding the queried words, which greatly improves the efficiency of the method by reducing the processing time.

## 3. Proposed method

In this section we present an overview of the method we developed for the detection of cyberbullying messages on the Internet. In the proposed method we extend the method proposed by Turney (2002) [10], which calculates the relevance of seed words with words contained in the input entries, in our case, entries from electronic bulletin board (BBS) pages. Turney's method is extended firstly in the sense that the relevance with a seed word is calculated not for simple words, but for phrases automatically extracted by using dependency relations. Secondly, the score is maximized for each of the applied category of seed words. We apply multiple categories of harmful words and calculate the degree of association separately for each category. Finally, as the harmfulness score (or "harmful polarity") we chose the maximum value from those achieved by all categories. The method consists of three general steps:

1. Phrase extraction,
2. Categorization and harmful word detection together with harmfulness polarity determination,
3. Relevance maximization.

Each of the steps is explained in detail in the following paragraphs.

### 3.1. Phrase extraction

For some explicit cases of cyberbullying entries the harmful character of an entry could be determined by looking at separate words, as it was demonstrated in previous research [27,9]. For other cases however, even if a word in itself is not harmful, it gains harmful meaning when used in a specific context, or in combination with other words. For example, for a pair of words 性格が悪い (*seikaku ga warui*) "bad personality", neither "bad", nor "personality" on their own express harmful meaning. However, when these words are used together in a dependency relation, they become harmful (negative depiction of a person's personality). Therefore, methods for detecting harmful contents using separate words only, will fail when they encounter an entry which has a harmful meaning for phrases containing words in a dependency relation.

To deal with this issue, we calculated the harmful polarity score not for single words, but for all combinations of morphemes[4] in dependency relations. We define such combinations as "phrases" (one phrase consists of a morpheme pair in a dependency relation). The dependency relation is calculated by using morphological information provided by a morphological analysis tool for Japanese

[4] In this paper, we use the words "word" and "morpheme" in the same meaning, as they are processed similarly by morphological analysis tools for Japanese language.

**Table 2**
Types of phrases applied in the proposed method with examples.

| Phrase | Example |
|---|---|
| Noun–noun | サル顔 (*sarugao*) "monkey face" →Description ridiculing person's features |
| Noun–verb | 新田を殺す (*Nitta wo korosu*) "Kill Nitta" →Threatening expressions |
| Noun–adjective | 性格が悪い (*seikaku ga warui*) "bad personality" →Description criticizing person's features |

(MeCab[5]) parsed by a dependency parser for Japanese (Cabocha[6]). The phrases defined this way are extracted from all target entries. The strength of this approach is that such basic language analysis tools (morphological analyzers and dependency parsers) exist for most common languages, which makes the proposed method language independent. Examples of such extracted phrases are represented in Table 2.

### 3.2. Harmful word detection and categorization

In this process we detect the words and phrases of potential bullying connotations, later referred to as "harmful words/phrases". Harmful words often include newly coined words [29] or informal modifications of normal transcriptions [28], thus are not recognized by standard preprocessing tools, such as morphological analyzers or dependency parsers. Therefore, it is possible such words would not be handled properly and would cause error in morphological analysis, unless specifically annotated. We investigated the entries of informal school Websites using the definition of harmful words proposed by the Japanese Ministry of Education, Culture, Sports, Science and Technology (later referred to as "Ministry of Education", or MEXT) [5], and registered 255 harmful words (nouns, adjectives and verbs) in the dictionary of a morphological analyzer.

In addition, we categorized harmful words into three categories: obscene, violent and abusive. In this regard we followed the definition provided by The Ministry of Education of Japan, which specifically mentions words frequently used in cyberbullying messages to include those three categories. In this study, we applied the above official definition as it follows the country's policy in handling of cyberbullying cases, so we also classified harmful words into the three mentioned categories.

Next, from each category we automatically selected as seed words three words most often occurring in the context of cyberbullying and registered them in the system. The words we selected included: セックス (*sekkusu*) "[to have] sex", ヤリマン (*yariman*) "slut", フェラ (*fera*) "fellatio/blowjob" for the obscene category, 死ね (*shine*) "die [imperative]", 殺す (*korosu*) "[to] kill", 殴る (*naguru*) "[to] slap" for the violent category, and うざい (*uzai*) "annoying", きもい (*kimoi*) "gross", 不細工 (*busaiku*) "ugly" for the abusive category.

[5] http://taku910.github.io/mecab/.
[6] http://taku910.github.io/cabocha/.

### 3.3. Maximization of relevance score

In this process we calculated a harmfulness polarity score of automatically extracted phrases 3.1 with each seed word for all three categories 3.2. To measure the semantic orientation we applied the Point-wise Mutual Information (PMI) score, which in our implementation measures relevance between a phrase and harmfulness polarity words from each category. PMI here indicates a co-occurrence frequency of the queried phrase with the three words registered for each category. To calculate the co-occurrence frequency we use an information retrieval (IR) score (co-occurrence hit rate). Since countless numbers of various Web pages exist on the Internet, and various words are written there, it is possible to obtain a high coverage of the method by using the IR score.

We calculate the relevance of a phrase with words from each category according to Eq. (1). In this equation $p_i$ is a phrase extracted from the entry, $w_j$ are the three words that were registered in one category of harmfulness polarity words, $hits(p_i)$ and $hits(w_j)$ are Web search hits for each category for $p_i$ and $w_j$ respectively, $hits(p_i\&w_j)$ is a number of hits when $p_i$ and $w_j$ appear on the same web page. Finally, $PMI - IR(p_i, w_j)$ is the relevance of $p_i$ and $w_j$.

$$PMI - IR(p_i, w_j) = log_2 \left\{ \frac{hits(p_i\&w_j)}{hits(p_i)hits(w_j)} \right\}. \quad (1)$$

From all three scores calculated for the phrase with seed words from the three categories, we selected the category which achieved the highest score as the one with the highest relevance with the phrase. We calculated the relevance score this way for all phrases extracted from the entry. Finally we selected the category with the maximal overall score as the one with the highest relevance with the entry. Thus the *score* was calculated according to the following Eq. (2).

$$score_{entry} = max_{all_phrases}(max_{all_c ategories}(PMI - IR(p_i, w_j))). \quad (2)$$

In previous research, which will be considered here as a baseline, Matsuba et al. (2011) [9] calculated the relevance as a sum of all scores for all phrases with each harmful word separately. In this method instead of taking all words separately and calculating the score as the sum of all intermediate PMI scores, we grouped them into categories and calculated the relevance with the three most common harmful words from each category. By incorporating this improvement during the Web search the retrieved pages are those for which the phrase appeared not only with at least one of the harmful words, but with all three words from one category. This reduces the processing time, and, what is more important, improves the calculation of the relevance score, since only the strongest (most harmful) phrases are selected. Moreover, since it is easier to find a Web page containing a phrase with only one harmful word than a phrase with three words together, this baseline method mistakenly allowed phrases with low actual relevance to bullying entries to achieve high scores. Maximization of the relevance *score* prevents low relevance phrases to erroneously achieve high scores.

We explain the above process with the following example: 可愛いけど性格が悪い女 (*Kawaii kedo seikaku ga warui onna*) "Cute girl, but bad personality". Firstly, from the entry the extracted phrases are: 可愛い - 女 (*kawaii-onna*) "cute-girl", 性格 -悪い (*seikaku-warui*) "bad-personality", 悪い -女 (*warui-onna*) "bad-girl". Next, we calculate the relevance between "cute-girl" and the three groups of words separately ("sex, slut, fellatio", "die, kill, slap", "annoying, gross, ugly"). The highest maximal score is selected as the relevance (harmfulness score) for this phrase. Similarly the score is calculated for "bad-personality" and "bad-girl". From all the scores for all phrases the highest overall score

is considered as the maximized harmfulness *score* of the entry. All entries are then sorted beginning with the one with the highest harmfulness score. Finally, we set a harmfulness threshold *n* and consider *n* entries with the highest score as harmful, and discard other as irrelevant to check how many of the entries within the specified threshold are in fact harmful.

## 4. Evaluation experiment

We performed an experiment to evaluate the performance of the proposed method, and compared the results with the baseline. Below we describe the dataset prepared for the experiment (Section 4.1), performed the preliminary study for carrying out the experiment (Section 4.2), explained the experiment settings (Section 4.3) and reported on the results of experiment (Section 4.4).

### 4.1. Dataset

At first we needed to prepare a dataset for the experiment. We used the dataset created originally by Matsuba et al. (2010) [32] and developed further by Matsuba et al. (2011) [9]. The dataset was also used by Ptaszynski et al. (2010) [27], Nitta et al. (2013) [11]. It contains 1490 harmful and 1508 non-harmful entries. The original data was provided by the Human Rights Research Institute Against All Forms for Discrimination and Racism in Mie Prefecture, Japan,[7] and contains data from unofficial school Web sites and forums (BBS). The harmful and non-harmful sentences were manually labeled by expert annotators who are Internet Patrol members, according to instructions included in the MEXT manual for dealing with cyberbullying [5]. Some of those instructions are explained shortly below.

The MEXT definition assumes that cyberbullying happens when a person is personally offended on the Web. This includes disclosing the person's name, personal information and other areas of privacy. Therefore, as the first feature distinguishable for cyberbullying MEXT defines private names. This includes such information as:

- Private names and surnames (e.g."Michal Ptaszynski"),
- Initials and nicknames (e.g. "Mr.P.", "Mi*al Ptasz*ski"),
- Names of institutions and affiliations (e.g. "That foreigner assistant professor from Kitami Institute of Technology").

As the second feature distinguishable for cyberbullying MEXT defines any other type of personal information. This includes:

- Address, phone numbers, etc. (e.g. "165 Koen-cho, Kitami, 090-8507, Japan", or "+81-157-26-9327"),
- Questions about private persons (e.g. "Who is that tall foreigner straying on Computer Science Dept. corridors?"),
- Entries revealing personal information (e.g. "I heard that guy is responsible for the new project".).

Literature on cyberbullying also indicates vulgarities as one of the most distinctive features of cyberbullying [12,8,27]. Also according to MEXT, vulgar language is distinguishable for cyberbullying, due to its ability to convey offenses against particular persons. Examples of such words are, in English: *shit*, *fuck*, or *bitch*, in Japanese: *uzai* (freaking annoying), or *kimoi* (freaking ugly). In particular MEXT distinguishes three kinds of vulgar or inappropriate language, namely obscene, violent and abusive. We incorporated these categories in our research.

In the prepared dataset all entries containing any of the above information was classified as harmful.

[7] http://www.pref.mie.lg.jp/jinkenc/hp/.

### 4.2. Preliminary study

After preparing the initial datasets for experiments, it was necessary to create test data with harmful and non-harmful entries mixed at an appropriate rate. In previous research the mixing was set at the same rate (half of the data contained cyberbullying entries). However, it cannot be assumed that harmful entries appear in real life with the same rate as normal ones. Therefore to evaluate our method in conditions closer to reality we performed a preliminary study to verify how much of the entries are harmful on actual Web pages. We counted the harmful entries mixing ratio on three informal school Websites, in particular we focused on informal school bulletin boards (BBS). The result of the study is represented in Table 3. We performed the study during four days between January 27 and 30, 2012. The number of obtained entries was 2222.

As of the result of the study, the first BBS contained a total number of 600 entries from which 75 were harmful, which indicates that harmful entry appearance rate was 12.5%. Similarly, for the second BBS, 90 out of 736 total entries were harmful (12.2%). On the third BBS there were 886 total entries with 100 harmful ones (11.3%). From the above results, we concluded that about 12% of all entries appearing on informal school BBS can be accounted as cyberbullying. Therefore in the experiment we verified the performance of the method under the condition when harmful entries cover 12% of the whole data.

### 4.3. Experiment settings

In the evaluation experiment we compared the performance of the proposed method to the baseline [9]. We did this firstly for the case where the test data contained 50% of harmful entries, to first verify the performance under the same conditions as in the previous research. Next, we prepared different test data, which contained 12% of harmful entries and compared the performance under this condition for both the baseline and the proposed method.

Firstly, on the prepared initial dataset (see Sections 4.1 and 4.2) we performed a 10-fold cross validation. We processed the dataset by both the baseline and the proposed method and calculated the harmful polarity score for entries where the phrases could be extracted. Then we ranked all entries decreasingly according to the harmful polarity score, and evaluated the performance looking at the top *n* entries and increased the threshold *n* by 50 entries each time.

Next, we prepared the datasets for the verification in real world conditions (12% of harmful entries). We prepared five test sets by randomly extracting 60 harmful and 440 non-harmful, 500 entries in total from the original dataset. On these datasets we did not perform a 10-fold cross validation, since it would make the results not statistically relevant (each set for cross validation would contain only six (one tenth) of harmful entries). Instead, we calculated the results for each of the five sets separately. This allowed us to include all entries from the original test set in the evaluation. We processed these datasets with both the baseline and the proposed method and calculated the harmfulness polarity score for entries for which the phrases could be extracted. Then, similarly to the original dataset, we ranked all entries based on the harmfulness polarity score, and evaluated the performance by taking the top *n* and increasing the threshold *n* by 10 entries each time.

We considered an automatic setting of the threshold using machine learning methods, however it was difficult due to the small size of test data for the real world conditions. In the future for automatic threshold setting we plan to develop a machine learning method capable of handling small sized and sparse data. This time

**Table 3**
The numbers of entries on informal school BBS including the number and percentage of cyberbullying entries.

| BBS | Overall number of entries | Cyberbullying entries | Percentage (%) |
|---|---|---|---|
| BBS(1) | 600 | 75 | 12.5 |
| BBS(2) | 736 | 90 | 12.2 |
| BBS(3) | 886 | 100 | 11.3 |

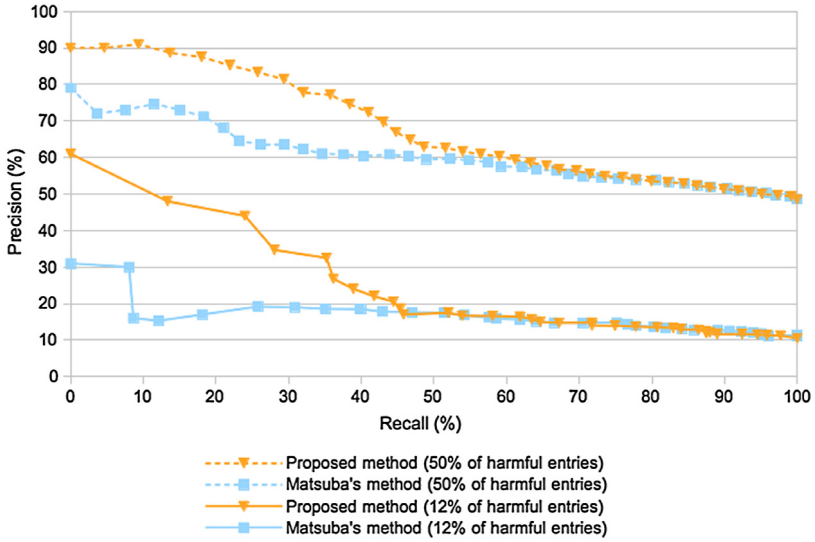



**Fig. 1.** Precision and Recall for both the baseline and the proposed method for both datasets (50% and 12% of cyberbullying entries). The horizontal axis and the vertical axis represent percentage of Recall and Precision for each threshold, respectively. All results were statistically significant. For the 50% data the results were extremely significant on 0.0001 level. For the 12% data the results were statistically significant on 0.05 level.

we increased the threshold manually each time and investigated Precision and Recall for each threshold.

As evaluation criteria we used Precision (P) and Recall (R), calculated according to the Eqs. (3) and (4). P and R are standard evaluation measures defined in 1955 [33], and from that time used as standard evaluation measures in many fields, including mathematics, statistics, and more recently information retrieval. Precision is traditionally defined as a ratio of all correctly retrieved documents to all retrieved documents. Recall on the other hand is the ratio of the all correctly retrieved documents to all correct documents available.

Within the context of present research, Precision is a ratio of the number of harmful entries that were correctly determined as harmful by the proposed method to the number of all entries determined by the system as harmful among the top $n$ cases. For example, when the proposed retrieves 50 documents (in other words classifies 50 documents as harmful), not all of them are classified correctly, some of them could be assigned a high harmfulness score mistakenly. If, for example 45 out of 50 entries are classified correctly, Precision is equal to 90% (45/50).

Recall in our research is understood as a ratio of the number of entries that could be correctly determined by the proposed method as harmful to the overall number of harmful entries in the dataset (1490). Therefore, if for the first threshold window (top 50 entries with the highest assigned harmfulness score) all are classified correctly, the Recall for this threshold window is about 3.4% (50/1490). However, if for the first threshold window, only 45 entries are harmful and the remaining five entries are mistakenly classified non-harmful entries, the Recall is equal to about 3% (45/1490).

The final performance is calculated as a balanced mean of Recall and Precision for each test data in this experiment. As an additional evaluation mean we also looked at Area Under the Curve (AUC) for graph representing Precision and Recall on the $y$ and the $x$ axis respectively.

$$P = \frac{correct\ \ annotations}{all\ \ system\ \ annotations} \tag{3}$$

$$R = \frac{correct\ \ annotations}{all\ \ harmful\ \ entries}. \tag{4}$$

### 4.4. Results

The results showing Precision and Recall for both the baseline and the proposed method for both datasets (50% and 12% of cyberbullying entries) are represented in Fig. 1. The horizontal axis and the vertical axis represent percentage of Recall and Precision for each threshold, respectively.

For the test data containing 50% of harmful entries, Precision was between 49% and 79%, with Recall oscillating between 3% and 100% for the baseline, while for the proposed method Precision was between 49% and 88%, with Recall between 5% and 100%. For the test data containing 12% of harmful entries, Precision was between 11% and 31%, with Recall oscillating between 8% and 100% for the baseline, while for the proposed method Precision was between 10% and 61%, and Recall was between 13% and 100%. This suggests that the proposed method outperformed the baseline. AUC also indicated the same result. The results are further explained and discussed in detail in Section 4.5.

### 4.5. Discussion

The experiment results showed that the proposed method achieved higher overall performance compared to the baseline. The shape of the correlation curve for Recall and Precision shows that the performance for the baseline was significantly lower in general comparing the test data containing 12% of harmful entries to the test data containing 50% of harmful entries. However, for the

proposed method, although the performance is reduced as well, there was no sudden drop in the shape of the correlation curve when comparing both datasets.

This could suggest that the performance was more stable in the proposed method than in the baseline. There were several cases of threshold *n* where the Recall was slightly higher for the baseline than for the proposed method in the test data containing 12% of harmful entries. This happened because for some harmful entries the harmfulness score could not be calculated highly enough due to the fact that the score calculation is more strict (Precision-oriented) in the proposed method. Therefore, although Recall is slightly higher in the baseline for high thresholds, it is always still higher for the proposed method for small and medium thresholds, with the Precision being constantly higher for the proposed method. Therefore, it can be concluded that the proposed method achieved higher general performance than the baseline.

Next we explain the results for the test data containing 12% of harmful entries. We investigated the threshold cases of entries where the Precision reaches 48%. Entries found there included, for example "アトピーのやつ死ねよ" (*atopii no yatsu shine yo*) "The bastard with atopy must die" and "ウザイキモイぶす" (*uzai kimoi busu*) "annoying, gross and ugly". From those entries high relevance score was calculated for phrases like "アトピー-死ね" (*atopii-shine*) "atopy-die" and "ウザイ -ぶす" (*uzai-busu*) "annoying-ugly". Since the phrases contained seed words as well, this most probably increased the polarity score of the harmful entry.

On the other hand, there were many non-harmful entries incorrectly classified as harmful with harmfulness polarity score equally high or higher than the actual harmful entries. An example of a non-harmful entry of this kind was "県外に住んでいる" (*Kengai ni sun de iru*) "Living outside of the prefecture". The phrase extracted from this entry was "外 -住ん" (*soto-sun*) "outside-live". This is a neutral phrase, and appears often in non-harmful entries. However, it also appears in harmful entries in cases of exposing personal information about where a person lives. Therefore, the relevance of a harmful entry containing such a phrase is increased, and as a result the overall harmfulness polarity score of such non-harmful entry becomes higher.

As a countermeasure it could be considered to register some words, which appear only in non-harmful entries, like "beautiful" or "splendid", with non-harmful polarity and calculate the relevance of non-harmful entries as well. In particular, firstly, the non-harmful words could be registered in the dictionary. Then the relevance could be calculated for the phrases with both, non-harmful polarity words and harmful polarity words. In such cases the phrase could be considered as non-harmful when the relevance score of the phrase with non-harmful words was higher than the relevance with the harmful words. This could reduce the influence of neutral phrases on the overall performance.

We also investigated the cases where Recall reaches 100%. These cases include entries which contain personal information only such as the school name or a person's name, such as "Nitta of Kitami Institute of Technology, 4th grader" (name changed only to illustrate the example), etc. The relevance of such entries with harmful words registered at present in the dictionary is low, which influenced their overall harmfulness score as well. To solve this problem we plan to register words in the dictionary which have high relevance with personal information and use them in the relevance score calculation as well.

## 5. Deployment and re-evaluation of the method

After confirming the performance, the proposed method was deployed as a working system and tested "in the field" for over one year. It was used to analyze informal electronic forums and BBS of various universities including Kitami Institute of Technology (which the authors belong to), Asahikawa Medical University, Otaru University of Commerce, and Obihiro University of Agriculture and Veterinary Medicine. Despite some successes at first, after some time we started to notice that fewer and fewer entries were picked up by the system each day. Therefore we decided to perform a re-evaluation of the system under the same conditions we evaluated in 2013. In the re-evaluation we also applied the same metrics we used previously, namely Area Under the Curve (AUC) on the graph showing Precision and Recall.

After re-evaluating the method after one and a half years we found out that the performance changed since the time of its development. The performance significantly dropped, losing over 30-percentage-points compared to its original performance. The result of the comparison is represented in Fig. 2.

### 5.1. Discussion on reasons for performance drop

In this section we discuss the possible reasons for the decrease of performance in the proposed method.

After a thorough analysis of the experiment data we noticed that most of the information extracted in 2013 was not available in 2015. This could be due to the following reasons. Firstly, fluctuation in page rankings could push the information lower making it unextractable by the proposed method. Secondly, frequent deletion requests of harmful contents by Internet Patrol members could have had an effect as well. However, the most probable is the third cause, which is the recent tightening of usage policies by most Web service providers, such as Google,[8] Twitter[9] and *Yahoo!*, which are used in the proposed method in the Information Retrieval procedure (Section 3.2). This also includes recently introduced DMARC[10] policies related to e-mail spoofing and general improvements in policies aimed at decreasing Internet harassment. However, the fact that the performance decreased from over 90% to less than 60% during less than two years is an important warning also for other research based on Web mining. Probability of such problems have been indicated some time ago by Kilgarriff (2007) [34], and could become a major problem in the future. This also advocates more focus on either corpus-based methods or methods providing optimization and constant update of applicable seed words (training features). In the following sections we focused on the latter, since it allows improving the method even when the performance drops again in the future.

## 6. Automatic seed word acquisition and filtering

In this section we explain our re-implementation and performance improvement of the proposed method for cyberbullying detection. To improve the lost performance we firstly developed a method for automatic acquisition of seed words which we combined with optimization method based on single- and double-filtering of the seed words.

### 6.1. Automatic acquisition of harmful words

To improve the performance of our method we first applied a method proposed by Ishizaka et al. (2011) [35].

The method proposed by Ishizaka et al. extends the initial value based on n-gram statistics and uses it to automatically acquire harmful expressions from a large corpus (for example the Web).

[8] https://www.google.com/events/policy/anti-harassmentpolicy.html.
[9] https://blog.twitter.com/2014/building-a-safer-twitter.
[10] http://www.dmarc.org/.

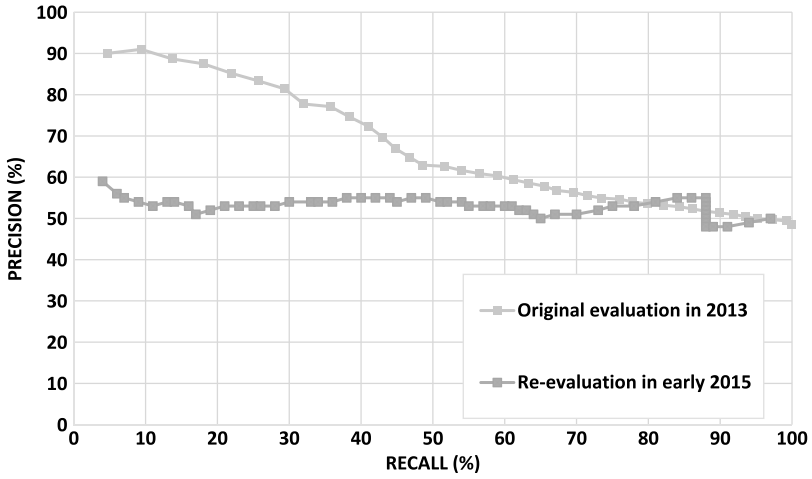



**Fig. 2.** Performance comparison for the proposed method in late 2013 and re-evaluated in early 2015. The evaluation performed for fair condition with data balanced for harmful and non-harmful entries.

1. J: *Busaiku dashi uta wa heta dashi dansu mo heta.*
   E: She's ugly, sings badly, and cannot even dance.
2. J: *Omake ni DQN darake.*
   E: Apart form that all moron.
3. J: *Kumamon ni kuwasero, baka dono!*
   E: Feed this to Kumamon [TV character], you ass*ole!
4. J: *Ramen mania tte shinsoko kimoi*
   E: That Ramen maniac is like completely gross
5. J: *Munou dakara darou.*
   E: That's 'cause his an impotent.

**Fig. 3.** Examples of extracted harmful statements. Japanese romanized (**J**), English translation (**E**, adjusted to express the original meaning as closely as possible).

It calculates the SO-PMI-IR value (later generalized as "SO value") indicating the degree of harmfulness of a word in a document (for explanation of SO-PMI-IR see Eq. (1) in Section 3.3). The difference in our method is that Ishizaka et al. focuses on optimizing the relevance between words, not phrases. In particular, after obtaining an initial set of words (for example harmful and non-harmful), they calculate SO value between all words in the set to provide clusters of words being in close relevance. Originally, the method was successfully applied to harmful word acquisition from a large Japanese electronic bulletin board system (BBS), called "2 channel".[11] Some of the examples of harmful statements extracted with this method are represented in Fig. 3. All example sentences contain harmful expressions.

The method of Ishizaka et al. allows modification and updating of the initial value in the proposed method, which consists of a set of harmful seed words used in the IR procedure (Web mining). Such words could be automatically acquired and updated regularly, which would greatly decrease the costs of collecting the seed word lists when the Web mining conditions change. We assumed this method could be used to optimize the seed word set used in the proposed method.

### 6.2. Double filtering of harmful seed words

We conducted an experiment using Ishizaka et al.'s method to verify its potential in the automatic selection of seed word candidates for the developed cyberbullying detection method.

Ishizaka et al. in their study at first conducted a questionnaire in which they asked three annotators to judge whether a word is harmful or not. In the questionnaire they used a set of 2735 words automatically extracted from harmful BBS entries. From the initial set they retained only those words for which all three annotators agreed. There were 76 such words (we refer to them as harmful word candidates). These words were used as a base for automatic selection.

Next, Ishizaka et al. automatically selected a set of 14 most frequent harmful and 17 non-harmful basic words appearing in the original entries by using n-gram statistics (with $n = 7$). They used those basic words to filterout the most harmful of the initial 76 harmful word candidates selected by human annotators. In our approach we added to the Ishizaka et al.'s 14 harmful basic words 9 harmful seed words used by Nitta et al. With 23 harmful and 17 non-harmful basic words we conducted the first filtering of the basic words. We calculated SO values for each of the 76 harmful word candidates with every other basic word, and calculated the sum of SO values for each word candidate. From the words which obtained SO value higher than 0 we selected only those words which also appeared in the group of basic words. The basic word set used in filtering is represented in Table 4. The basic words that were filtered out are represented in Table 5. This first filtering provided a set of basic harmful words which were both harmful from the point of view of human annotators and were in the strongest relevance to harmful BBS entries.

In the second filtering we used as basic words the 17 harmful basic words obtained in the first filtering and the same 17 non-harmful basic words (later called "seed17"), and calculated the SO value again, this time for a dataset containing 76 harmful word candidates and 17 non-harmful basic words (93 words in total). We added the non-harmful words to verify whether the harmful basic words cause a bias in scoring. If it did, the non-harmful words would also achieve a high SO value. For a comparison, instead of the filtered out 17 basic words, we also used only the original set of 9 harmful seed words used in the originally developed method (later called "seed9"), and similarly calculated the SO value for the whole dataset (93 words). Having all SO values, we calculated arithmetic mean of SO values $-\mu$ and standard deviation $-\sigma$, and automatically set a threshold $\alpha_1$ as $\mu \pm \sigma$. Finally, we extracted harmful word candidates $w$ and non-harmful word candidates $h$ as $h \leq \alpha_1 \leq w$. A graph visualizing this operation is represented in Fig. 4. All words extracted this way for sets *seed17* and *seed9* are represented in Table 6. Although one would expect the basic

[11] https://www.2ch.net/.

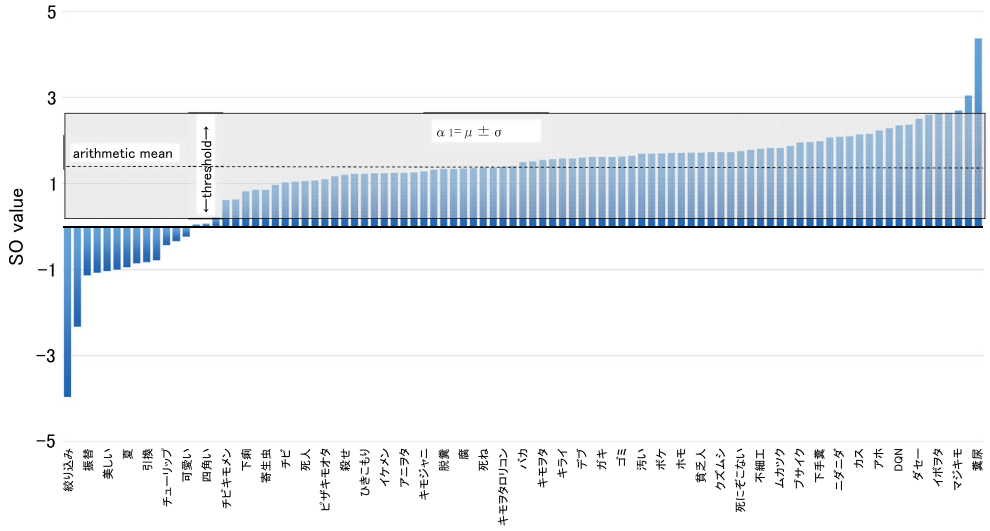



**Fig. 4.** Graph visualizing the procedure for extraction of word candidates for set *seed17*. All words with their translations and achieved SO values were represented in the Appendix at the end of the paper.

**Table 4**
Harmful and non-harmful basic words used in the experiment. English translations added by the authors to represent a meaning as close to original as possible. Capital letters indicate different characters used to write a word.

| | Harmful basic word candidates | Non-harmful basic word candidates |
|---|---|---|
| Proposed Ishizaka et al. [35] | *kiero* (get lost), *ujimushi* (worm), *kasu* (scum), *shineyo* (just die!), *dōtei* (virgin), *UZAi* (ANNOYing), *KIMOi* (UGLy), *kichigai* (f*cking crazy), *chon* (f*ckin' Chinese), *kuzu* (trash), *munō* (impotent), *BUSAIKU* (ugly), *busu* (ugly hag), *kirai* (hate), | *hikikae* (exchange), *kaiage* (purchase), *shiborikomi* (narrowing), *kawaii* (cute), *furikae* (transfer), *kōjun* (descending), *suteki* (nice), *ikemen* (handsome), *akai* (red), *subarashkī* (wonderful), *utsukushkī* (beautiful), *tsukue* (desk), *suppai* (sour), |
| | *sekkusu* (sex), *yariman* (slut), *fera* (fellatio, bl*wjob), *shine* (die), *korosu* (kill), *naguru* (beat the crap out of somebody), *uzai* (annoying), *kimoi* (gross), *busaiku* (ugly) | *chūrippu* (tulip), *taiyō* (sun), *natsu* (summer), *shikakui* (square) |

**Table 5**
Harmful 17 basic words with SO value ≥0.

| |
|---|
| *kiero* (get lost), *shineyo* (just die!), *munō* (impotent), *busaiku* (ugly), *busu* (ugly hag), *dōtei* (virgin), *chon* (f*ckin' Chinese), *BUSAIKU* (ugly), *shine* (die), *uzai* (annoying), *UZAi* (ANNOYing), *KIMOi* (UGLy), *kichigai* (fucking crazy), *kimoi* (ugly), *kasu* (scum), *korosu* (kill), *kuzu* (trash), |

**Table 6**
Harmful and non-harmful seed words obtained according to the double filtering procedure (*seed17*) and single filtering procedure (with original seed words used in the final filtering (*seed9*)).

| | Harmful seed words | Non-harmful seed words |
|---|---|---|
| *seed17* | *dasee* (hopeless), *bakasayo* (stupid communist), *fun-nyo* (dung), *majikimo* (seriously gross), *ibo-ota* (warty nerd), *kuzumasugomi* (shitty mass-mudia), *goki-ota* (cockroachy nerd) | *shiborikomi* (narrowing), *kaiage* (purchase), *furikae* (transfer), *kōjun* (descending), *natsu* (summer), *suteki* (nice), *utsukushkī* (beautiful), *kawaii* (cute), *hikikae* (exchange), *akai* (red), *taiyō* (sun), *chūrippu* (tulip), *tsukue* (desk) |
| *seed9* | *bicchi* (bitch), *ibo-ota* (warty nerd), *mekuso* (eye gum), *dappun* (feces), *kusomushi* (dung beetle), *bakasayo* (stupid communist), *dasee* (hopeless), *fun-nyo* (dung), *goki-ota* (cockroachy nerd), *maji-kimo* (seriously gross), *kuzumasugomi* (shitty mass-mudia), *gumin* (ignorant) | *shiborikomi* (narrowing), *chūrippu* (tulip), *kaiage* (purchase), *suteki* (nice), *taiyō* (sun), *tsukue* (desk), *natsu* (summer), *akai* (red), *shikakui*, *kōjun* (descending), *kawaii* (cute), *hikikae* (exchange), *furikae* (transfer), kiseichū (parasite) |

words to come on top, since they would be expected to obtain a higher SO value due to perfect relevance score with themselves, in the actual results the candidate words extracted according to this procedure did not contain even one word from the basic harmful word set. Instead, the words which were in the highest general relevance at the time (present Web condition) with the words used for optimization appeared on top. This indicates that the filtering optimized the search and can be considered as a useful procedure.

**Table 7**
Description of six seed word sets (*case1*–*case6*) applied in the experiment with the proposed cyberbullying detection method.

| | |
|---|---|
| *case*1 | 7 words from *seed17* (Table 6) |
| *case*2 | 12 words from *seed9* (Table 6) |
| *case*3 | 16 words (7 words from *seed17* + original 9 words) |
| *case*4 | 21 words (12 words from *seed9* + original 9 words) |
| *case*5 | 5 words originally used by Ishizaka et al. [35] |
| *case*6 | 9 words originally used in the developed method |

## 7. Optimization experiment and results

After obtaining seed word sets described in Section 6, we conducted an experiment to verify whether and to what extent does the change in the selection of seed words affect the originally developed method. Moreover, we calculated the statistical significance of the results using the McNemar Test [36].

### 7.1. Experiment setup

The experiment was designed as follows. Using the filtered out seed words we prepared six different harmful word candidate sets (later referred to as "cases"), applied them as seed words in the originally developed method and verified its performance. *Case1* contained 7 double-filtered harmful words (all harmful words from *seed17* in Table 6). *Case2* contained 12 harmful words obtained by single-filtering out the words with the highest harmful relevance using only the original seed word set (all harmful words from *seed9* in Table 6). *Case3* contained the original 9 seed words and 7 double-filtered harmful word candidates (overall 16 words). *Case4* contained the original 9 seed words and 12 single-filtered candidates. *Case5* contained 5 words originally used by Ishizaka et al. in their paper [35]. Finally, *case6* contained only the original 9 seed words without any modification or optimization. The cases were summarized in Table 7.

We used each case as a seed word set in the proposed cyberbullying detection method and conducted the analysis using the original dataset of 2998 entries 4.1. We decided to use the whole set, not the set containing 12% of harmful entries to verify how each change in the seed word set influenced the results in a fair balanced condition. Moreover, as the original experiment from Section 4 showed, the results scaled similarly for both kinds of evaluation datasets. Thus the differences between the change in performance depending on different cases remained similar. Also, when more than two cases were compared, a smaller and less-balanced dataset caused difficulties in finding valid differences between the cases for higher thresholds, and, most importantly, in testing the statistical significance of the results.

In the evaluation we looked at several metrics to choose the best seed word set. We calculated Precision (P), Recall (R) and balanced F-score (F) for each cases. Then we first (1) checked which case obtained the highest F-score for the longest threshold span. Next, (2) we looked at which case obtained the highest break even point (BEP) on a plot with P and R. Thirdly, (3) we looked at which of the cases obtained the highest Precision in general. Finally, (4) as the last mean of evaluation we used the same metrics as previously, namely, area under the curve (AUC) on a plot with P and R on $y$ and $x$ axis respectively. We also asked ourselves a question (5) whether it was always better to simply add more words, or whether the automatic filtering of harmful word candidates improved the method and to what extent.

### 7.2. Results

Below we address the five questions from Section 7.1 one by one.

**Table 8**
McNemar Test results (for *cases* 5 and 6 as baselines), $*p \leq 0.5$, $**p \leq 0.1$, $***p \leq 0.01$.

| | *case*1 | *case*2 | *case*3 | *case*4 | *case*5 | *case*6 |
|---|---|---|---|---|---|---|
| *case*5 | 189.00 *** | 26.88 *** | 0.83 | 0.30 | – | 16.98 *** |
| *case*6 | 145.00 *** | 5.80 * | 9.47 ** | 10.29 ** | 18.56 *** | – |

(1) The highest F-score in general was obtained by *case2* (68%). Also *case5* and *case6* obtained a similarly high F-score once, however, for most of the threshold, F for *case2* was always the highest, while for *case5* and *case6* mostly the lowest.
(2) All cases obtained similar BEP, with cases 1, 2 and 6 = 54% and cases 3, 4, 5 = 53%. Therefore BEP did not appear sufficiently informative, although *case2* being in the higher group indicated a minor support for the result from (1), where *case2* appeared the best.
(3) The highest Precision was obtained by *case4* (80%). However, despite obtaining the highest score once, Precision in case4 quickly decreased, ending with one of the worst results in general. The second best P was obtained by *case2* (76%). Moreover, *case2* retained high P for most of the threshold span. This supports the previous result indicating *case2*as the best choice.
(4) When it comes to AUC, *case2* outperformed all other cases for most of the threshold span. We consider this as a final support for *case2* obtaining the best performance. The results are represented in Fig. 5.

### 7.3. Discussion of the experiment results

Finally, we verified whether simply adding words to the original method improves its performance. After analyzing all evaluation metrics, the cases' performances were in order: case2 (12 words), case1 (7 words), case3 (16 words), case4 (21 words), case6 (9 words), case5 (5 words). This suggests the following conclusions. Simply adding words does not improve the performance. Moreover, the two best scores were obtained by the cases applying the proposed automatic acquisition and optimization method. Here, *case1* applies the full filtering procedure (*seed17*). *Case2* applies single filtering with a final set of harmful seed words extracted with the help of the original seed word set instead of the double-filtered word set. This suggests that the filtering improves the performance of the cyberbullying detection method in general. Moreover, shortening the filtering from double- to single-filtering positively influences not only the performance, but also efficiency of the method. Applying only single-filtering instead of double-filtering saves much of the processing time needed to extract the final set of words.

### 7.4. Statistical significance with McNemar test

We verified the statistical significance of the results using the McNemar Test [36], which is a standard test used for calculating statistical significance for paired nominal data. We considered *case5* and *case6* as baselines and calculated the significance for all other cases.

All cases except 3 and 4 compared to *case5*-baseline were statistically significant on at least the 5% level. The results were always extremely significant for *case1*, which we consider as the one with second-best performance. The results of the best performing *case2* were significant when compared to baseline-*case6* and extremely significant when compared to baseline-*case5*. The results of McNemar test were represented in Table 8.

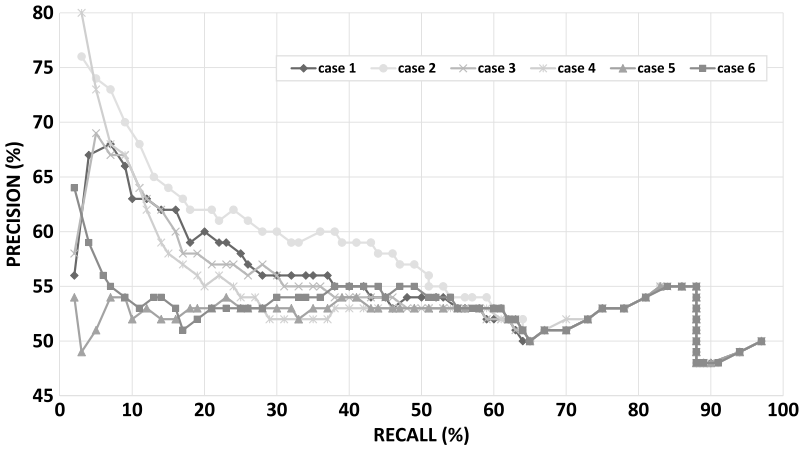



**Fig. 5.** Results of all compared cases.

## 8. General discussion on automatic cyberbullying detection tools

In this section we address some practical issues, regarding actual implementation of the proposed method in a larger cyberbullying prevention environment, that were not addressed fully. This includes not only the potential, but also limitations, as well as ethical issues regarding the use of such detection methods.

### *8.1. Potential and limitations*

Apart from our attempt to automatize the net-patrol process, there have been a number of attempts to deal with cyberbullying with the use of Artificial Intelligence, Machine Learning, or Natural Language Processing techniques. Some of them have been described in Section 2.2. However, most of the research is still in a developmental phase and has yet to be applied in practice. Although there are differences in details how each of those methods work, the main function of such methods, including the one proposed in this paper is detecting cyberbullying entries in written text. Even assuming, that such a method would be able to detect cyberbullying entries with perfect accuracy, which, as we could see in this paper is not a trivial problem, this simple function alone would not be of any use if not applied in practice, in a wider milieu of solutions. Several examples of such possible practical applications have been listed below.

(I) Automated Internet crawler for the use by Internet patrol members to minimize their workload, and psychological burden.
(II) Software for monitoring one's own and other people's (friends, one's children) accounts to search for bullying-related contents.
(III) Gamification-based solution, in which users are engaged in a simple social game, such as conversation with a group of peers and a non-playable character avatar, during which the users are made to deal with cyberbullying situations.
(IV) Policy-based implementation, where an Internet portal or SNS applies the method to automatically spot and inform the user that harmful contents were found in their online post.
(V) Local solution for computers or mobile devices to detect and worn a user that the contents they have just typed contains harmful words or word patterns.

When it comes to the most straight-forward application, we developed our method for use in solution I. However, potentially the method could be applied to any of the above-mentioned possible applications. As for the examples of the above solutions, the authors found the following software either in development or available for use already.

Application type II. is realized by such software as **Samaritans Radar**. It was launched on 29 October 2014, by an organization focused on suicide prevention, called "Samaritans".[12] It was a free Internet application for Twitter, which helped users monitors their friends' Tweets. The main function of the application was alerting a user when it spotted anyone in the user's online surroundings, who could be either bullied, depressed or sending disturbing suicidal signals. Unfortunately, due to serious data protection and privacy issues, which were noticed by the users soon after the time of the launch, the application was closed permanently on 10 March 2015, six months after its release.

Another example of this kind of software is **PocketGuardian.** It is a newly released (September 2015) software for parental monitoring, that "detects cyberbullying, sexting, and explicit images on children's mobile devices".[13] By using machine learning techniques the software provides a statistical probability that the content (sentence, tweet, e-mail, or image) is inappropriate. An advantage of this software is that it focuses not only on textual content, but includes in its monitoring range the ability to detect explicit images. As for the disadvantage, the software is aimed at parents trying to monitor their children's mobile devices, which raises questions regarding the ethics of using such software and its influence on parent–child trust relationship.

Application type III. is realized by such software as **FearNot!**[14] [37]. Its developers describe it as "an interactive drama/video game that teaches children strategies to prevent bullying and social exclusion". The software uses psychology-inspired character AI which engages with users and teaches them how to deal with cyberbullying situations in real life. Unfortunately, the development of the software was stopped in early 2013.

Another example of the gamification-based approach to the problem is represented by the recently presented **StopBully**,[15] a serious game developed by the Department of Informatics at the Faculty of Sciences of the University of Lisbon [38]. The main goal of the game is to engage users in various real-life-like situations to teach them empathetic behaviors and through such positive behaviors prevent bullying in real life.

12 http://www.samaritans.org/.
13 https://gopocketguardian.com/.
14 http://sourceforge.net/projects/fearnot/.
15 http://www.di.fc.ul.pt/~apc/StopBULLY/.

Application type IV. is represented by, for example, **the New Twitter Policy.** On February 26, 2015, Twitter changed its policy regarding safety and misbehavior among its users,[16] in which they allowed users to report particular tweets as harassment incidents. An account of a confirmed bully becomes locked and can be reopened only under the condition that the bully deletes her harmful tweets. This is not yet a software *per se*, and for the time being Twitter simply declared increasing the number of staff in their support team focused on handling abuse reports. However, they claim to plan applying automatic approaches in the future as well.

Application type V. is realized by such software as **ReThink**.[17] It is an application for smartphones, which shows a pop-up warning message when a user tries to send a message containing harmful contents. Unfortunately, the algorithm for the detection of harmful contents in ReThink is based on simple keystroke logging and detecting vulgar and harmful words within the string of characters. This makes it inapplicable for more sophisticated contents, not including vulgar expressions. It also fails when user makes a mistake during writing and e.g., uses a backspace, since the "backspace" character is also recorded and hinders the detection of harmful words. Moreover using the keylogger function is also questionable. This way the software can track any kind of contents the user inputs on the keyboard, including personal information such as account names or passwords.

Another example of the software realizing application type V was patented some time ago by Mausi et al. (2013, **Patent Application No. 2013-245813**) [39]. In their approach the user is also informed about possible harmfulness of a message he or she has typed. However, differently to ReThink Mausi et al. (2013) do not use keylogger function, but rather allow the user to consciously type a message in a prepared user interface to verify its possible level of harmfulness. This patented approach was recently applied in an Android application [40].

The disadvantage of most of the above software is that in most cases the applications are based on the simplest methods (keylogging, simple pattern matching on words). Although being a good example of a quick and ad-hoc response to the cyberbullying problem, their accuracy would be limited to simple, explicit cyberbullying examples. Therefore future research will need to combine the simplicity of the ad-hoc solutions with the depth of computational models, such as the one proposed in this paper.

Finally, the limitations all of the software mentioned above and methods described in Section 2.2, is the fact that text classification approaches would not be able to deal with other typical features of cyberbullying, such as online ostracism, impersonation (personality theft by taking over one's account) or "outing" (revealing a person's private information and secrets). Dealing with those features will require an analysis of a much wider context, than one input sentence. For example, the system would need to have access to users secrets to distinguish whether the "outed" information would be in fact harmful for the user. It would also need to have constant access to the user's contacts and monitor all online behavior. It is possible that dealing with all those issues would be difficult, or even impossible from the point of view of actual real-life application, as it would be too dangerous for one software to be granted those large privileges.

[16] https://blog.twitter.com/2015/update-on-user-safety-features.
[17] http://www.rethinkwords.com/.

### 8.2. Ethical issues

Regardless of the technological side of the developed system, when it comes to a problem of such a weight as cyberbullying, one must always take into account ethical implications resulting from the proposed solution. Below we briefly summarize potential ethical issues which could appear for each of the types of applications described in Section 8.1. Type (I) of the application implies that the software will be used by an expert, such as a net-patrol member. This makes this a safe approach because the final decision on whether the detected contents are malicious or not will be made by a trained human.

Presumably, the most dubious kind of application is type (II), where the system is allowed to monitor the user's behavior and their online environment in the search for cyberbullying-related contents. Privacy issues which result from such an approach have been serious enough to make one of the application (Samaritans Radar) close shortly after launch.

Gamification (type III), when realized in its basic form also does not indicate much ethical issues. Some issues, apart from those like user account management, could arise when the game in which the user participates is not monitored by an expert, but the user is rather left alone with the game contents.

Policy-based applications are presumably the most difficult to estimate. On the one hand, when the service in question (such as Twitter) introduces a new top down policy, the users are always informed about the changes and are (or at least should be) given the choice to either continue using the service, or disagree with the new policy and stop using it. On the other hand, when a large corporation, such as Facebook introduces a new policy with which the users do not agree, such as forced advertisements using the user account information for providing directed ads, the users, who have been using Facebook for several years and have a developed friend network, would in practice most probably be forced to tolerate the new policy.

Finally, when it comes to local solutions, it seems that ethical issues emerge depending on the application. For example, using a keylogger function, which in practice keeps record of all of the user's behavior, including various account names and passwords, could be considered equally or even more malicious than the second approach. However, when the analysis of the input is performed after the user's agreement, such a solution could become a useful one.

## 9. Conclusions and future work

In this study, we proposed a method of maximization of category relevance to automatically detect cyberbullying entries on the Internet. With this research we wish to contribute to reducing the burden of Internet patrol personnel who make efforts to manually detect harmful entries appearing on the Internet. In order to verify the actual usefulness of the proposed method we evaluated the performance for the test data containing similar percentage of harmful entries as in reality. Firstly, in a preliminary study we verified the usual ratio of harmful entries on the Internet. Next, we prepared test datasets containing the same amount of cyberbullying entries as in the real world and evaluated the method on these test sets. In addition, we reproduced the baseline method and compared the performance to the proposed method.

The experiment's results showed that the proposed method obtained higher results than the baseline. Under the fair condition (test dataset with 50% of harmful entries) the proposed method achieved over 90% of Precision at 10% Recall and keeping up high Precision (80%–70%) at Recall close to 50%. Under real world conditions (test dataset with 12% of harmful entries) the method achieved nearly 50% of Precision at about 10% of Recall. The

relevance curve decreased slowly with growing Recall for the proposed method, while for the baseline the relevance curve dropped suddenly from 30% to around 15% at the same Recall rate. As for drawbacks in the proposed method, harmful entries consisting of personal information were scored as less harmful due to the appearance of neutral phrases which appear often in both harmful and non-harmful entries.

After deploying and testing the system for over a year we re-evaluated it, and noticed a sudden drop in performance (over 30% points). The loss could have been caused by several factors, including tightening of usage policies by many Internet services. To deal with this problem we proposed a method for automatic acquisition and optimization of seed words based on the method by Ishizaka et al. [35], and performed an experiment on multiple automatically constructed seed word sets. The results clearly showed that simply adding more words to the original method is not effective. On the other hand, the proposed filtering method was significantly effective, although shortening the filtering from double to single improved not only efficiency of the method, but also its performance, regaining much of the originally achieved performance. Moreover, the optimization not only improved the proposed method for the present time, but it can also be used to improve it in the future, if the results drop again due to fluctuations in Web information.

In the near future, we plan to deal with the remaining drawbacks. These include registering non-harmful polarity words which have a high relevance with non-harmful entries. This would lower the overall harmfulness polarity score of non-harmful entries containing neutral phrases. We will also investigate a method for assessing the harmfulness score to entries including personal information. Furthermore, by continuous testing of the method in practice we plan to semi-automatically collect additional datasets, and determine the optimal threshold automatically by using machine learning methods. We also plan to repeat the experiments by Ishizaka et al. to eliminate any potential ambiguities in human judgments about the harmfulness of words. Finally, we plan to apply the proposed optimization method to other Web-mining based tasks and verify its potential and limitations in improving such methods in general.

Finally, we plan to implement the proposed method as a part of a working solution for cyberbullying prevention. Our technology-based method cannot deal with all of cyberbullying features alone, such as ostracism, impersonation or outing. And even if it was able to detect those by sacrificing a user's privacy, it would not be able to solve the problem. A thorough solution would necessary need to include human experts, such as psychologists and educators to help both cyberbullying victims and those who become bullies by choice or by chance. Only then, would a software program combined with a well-planned social program would bring the desired effects and contribute to solving the problem of cyberbullying.

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## Appendix

All words used in the optimization experiment, sorted in a descending order by the achieved SO-value.

| Japanese | Romanized | Translation | SO value |
|---|---|---|---|
| 糞尿 | *Fun'nyō* | Excreta | 4.37 |
| バカサヨ | *bakasayo* | Stupid | 3.04 |
| マジキモ | *majikimo* | Seriously gross | 2.69 |
| クズマスゴミ | *kuzumasugomi* | trash of mass-mudia | 2.64 |
| イボヲタ | *ibowota* | nerd with warts | 2.63 |
| ゴキヲタ | *gokiwota* | nerd-cockroach | 2.59 |
| ダセー | *dasÄ“* | the worst | 2.50 |
| 糞虫 | *kusomushi* | Dung beetle | 2.37 |
| DQN | *DQN* | hooligan | 2.35 |
| キチガイ | *kichigai* | crazy | 2.29 |
| アホ | *aho* | Stupid | 2.23 |
| キモオタニート | *kimootanīto* | ugly nerd | 2.15 |
| カス | *Kasu* | scum | 2.14 |
| キモ | *Kimo* | ugly | 2.10 |
| ニダニダ | *nidanida* | tickety-tick | 2.09 |
| 目糞 | *mekuso* | eye mucus | 2.07 |
| 下手糞 | *hetakuso* | freaking clumsy | 1.98 |
| 愚民 | *gumin* | Ignorant | 1.96 |
| ブサイク | *busaiku* | Ugly | 1.95 |
| マジキチ | *majikichi* | seriously crazy | 1.87 |
| チョン | *chon* | Chinese | 1.83 |
| ムカツク | *mukatsuku* | freaking mad | 1.83 |
| 不細工 | *busaiku* | ugly | 1.81 |
| 消えろ | *kiero* | Get outta here | 1.78 |
| 死にぞこない | *shinizokonai* | zombi | 1.75 |
| クソジジイ | *kusojijī* | Old prick | 1.73 |
| クズムシ | *kuzumushi* | shit beetle | 1.73 |
| ウザイ | *uzai* | freaking annoying | 1.73 |
| ヲタ | *wota* | nerd | 1.72 |
| 貧乏人 | *binbōnin* | poor | 1.72 |
| 無能 | *munō* | impotent | 1.71 |
| ホモ | *homo* | Homo | 1.71 |
| ボケ | *boke* | idiot | 1.70 |
| 汚い | *kitanai* | dirty | 1.69 |
| 蛆虫 | *ujimushi* | scum | 1.69 |
| ビッチ | *bitchi* | Bitch | 1.65 |
| ゴミ | *gomi* | trash | 1.63 |
| メンヘラー | *menherā* | psycho | 1.62 |
| ガキ | *gaki* | Brat | 1.62 |
| 失せろ | *usero* | beat it | 1.62 |
| デブ | *debu* | fat | 1.60 |
| キライ | *kirai* | Hate you | 1.58 |
| キモオタ | *kimōta* | ugly nerd | 1.58 |
| 大嫌い | *daikirai* | hate you so much | 1.56 |
| キモヲタ | *kimowota* | ugly nerd | 1.55 |
| オタ | *ota* | nerd | 1.51 |
| バカ | *baka* | Fool | 1.49 |
| ロリコン | *rorikon* | Pedophile | 1.40 |
| キモヲタロリコン | *kimowotarorikon* | ugly pedophile | 1.39 |
| 死ね | *shine* | Die | 1.37 |
| 逝け | *ike* | die | 1.37 |
| 無知 | *muchi* | Ignorant | 1.36 |
| 腐 | *kusa* | Rot | 1.35 |
| 脱糞 | *dappun* | Defecation | 1.34 |
| 馬鹿 | *baka* | fool | 1.34 |
| オタク | *otaku* | nerd | 1.32 |
| キモジャニ | *kimojani* | ugly preppie | 1.28 |
| 悪臭 | *akushū* | Stench | 1.26 |
| 蠅 | *hae* | fly | 1.25 |
| アニヲタ | *aniwota* | anime nerd | 1.25 |
| アニオタ | *aniota* | anime nerd | 1.24 |
| イケメン | *ikemen* | handsome | 1.24 |
| 変態 | *hentai* | pervie | 1.22 |
| ひきこもり | *hikikomori* | hikikomori | 1.22 |
| 殺せ | *korose* | kill him | 1.20 |
| キモイ | *kimoi* | ugly | 1.17 |

(*continued on next page*)

| Japanese | Romanized | Translation | SO value |
|---|---|---|---|
| ピザキモオタ | *pizakimoota* | ugly pizza face nerd | 1.10 |
| 奇形 | *kikei* | mutant | 1.07 |
| 死人 | *shinin* | Dead | 1.06 |
| 猿 | *saru* | monkey | 1.04 |
| チビ | *chibi* | Runt | 1.02 |
| ジャニヲタ | *janiwota* | preppy nerd | 0.97 |
| ヨゴレ | *yogore* | Dirt | 0.85 |
| 寄生虫 | *kiseichū* | Parasite | 0.85 |
| 下痢 | *geri* | diarrhea | 0.82 |
| ブタ | *buta* | Pig | 0.63 |
| チビキモメン | *chibikimomen* | ugly small prick | 0.62 |
| 酸っぱい | *suppai* | sour | 0.22 |
| 四角い | *shikakui* | square | 0.07 |
| 素晴らしい | *subarashī* | Great | 0.05 |
| 可愛い | *kawaii* | cute | −0.23 |
| 赤い | *akai* | red | −0.33 |
| チューリップ | *chūrippu* | Tulip | −0.43 |
| 太陽 | *taiyō* | sun | −0.78 |
| 引換 | *hikikae* | Exchange | −0.82 |
| 素敵 | *suteki* | Lovely | −0.85 |
| 夏 | *natsu* | summer | −0.94 |
| 机 | *tsukue* | desk | −1.00 |
| 美しい | *utsukushī* | beautiful | −1.03 |
| 降順 | *kōjun* | descending order | −1.07 |
| 振替 | *furikae* | transfer | −1.13 |
| 買い上げ | *kaiage* | Purchase | −2.32 |
| 絞り込み | *shiborikomi* | Refine | −3.95 |

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